

# AI in the Real World

Data, Power, & Society - Day 01

CMSE 492 (aka CMSE 101) • Fall 2026

Prof. Danny Caballero



# Welcome!

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This is not a coding class.

You will not learn to build AI systems.

**Instead, you will learn to evaluate, question, and push back on them.**

This class aims to help you become a **Critical Pragmatist** when it comes to AI.

# Why This CMSE 492 (aka 101)?

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By the time you graduate:

- **You will work with AI systems.** Probably with many of them.
- **You will make decisions about AI.** How to use it, when to trust it, whether it should be used at all.
- **Forms of AI will be a part of your everyday life.** From websites to software to vehicles to appliances.
- **Others will ask your opinion.** "Does this AI system seem fair to you?" "Should we deploy this?" "Do you trust it?"

**This course prepares you to answer those questions with evidence, not intuition.**

# Who are your instructors?

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## Prof. Danny Caballero (he/they)

Physics & Astronomy; CMSE; CREATE for STEM

- Lappan-Phillips Professor of Math and Science Education
- Co-direct two research labs (in Physics & Computational Science Education)
- BS Physics, Texas; MS & PhD Physics, Georgia Tech; Postdoc Physics Education, CU-Boulder
- Former custodian, copy shop manager, high school physics teacher in Atlanta Public Schools
- Labor organizer for the Union of Tenure System Faculty (UTSF-MSU)
- Starting my 14th year at MSU (**Go Green!**)



*If you are comfortable, please call me "Danny"; Prof. Caballero is fine, too.*

# Who are your instructors?

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## Jorge Martinez-Ortiz

### CMSE & Political Science

- 4th year PhD Student in CMSE and Political Science
- BS Physics, Baylor University
- International student from Zacatecas, Mexico
- Treasurer for the Comunidad Latinoamericana (CLA) graduate student org.

*You can call me “Joy” or “Jorge”.*



# A Note on the Material

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This course covers topics related to AI that may be:

- **Sensitive** (algorithmic discrimination, bias, surveillance)
- **Controversial** (regulation, job displacement, autonomy)
- **Uncertain** (AI's future role, long-term impacts)

It's also going to touch on issues that affect people differently whether that be **gender, race, ethnicity, class, disability, or immigration status**.

- We are all affected by AI and more connected to each other than we realize.
- Each of you has **different biases, experiences, and perspectives**. That includes your instructors.

**We will not all agree. That's completely okay.** But we will listen and hear with humanity.

# Our Shared Framework: DTPA

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We use **one vocabulary** to analyze everything.

## The Data-Tools-Practices-Actions Framework

Instead of each person bringing their own lens, we will strive to speak the same language.

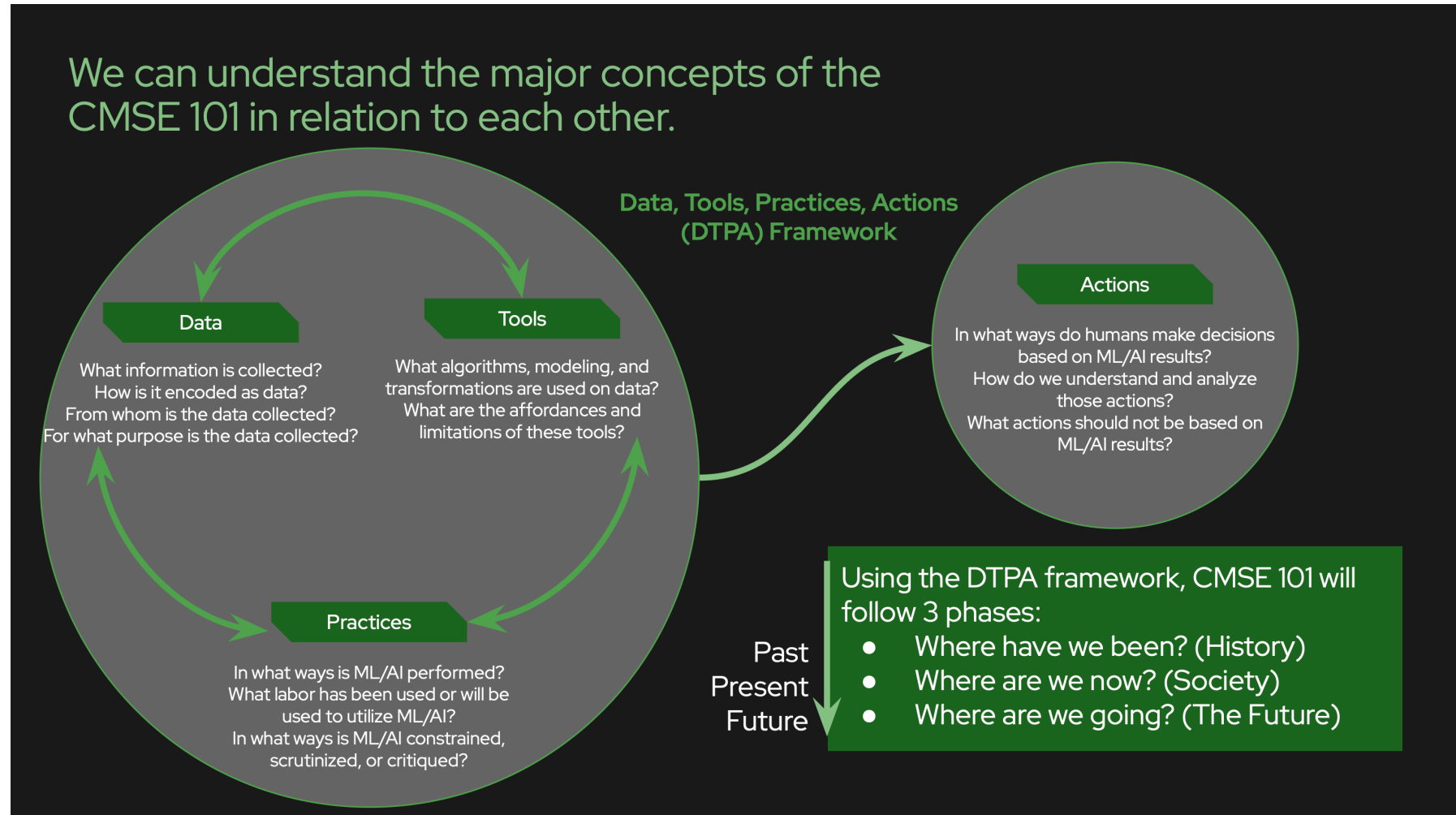
This lets us **disagree on conclusions** while **agreeing on facts**.

**By the end of the course, you'll be fluent in using DTPA for analysis.**

It can be your *starting* toolkit for evaluating *any* AI system.

*We also hope you will help us develop and strengthen this framework for teaching and learning about AI.*

# The DTPA Framework





# The Four Components of DTPA

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## **DATA**

What information is collected? Who is included/excluded? What gets erased?

## **TOOLS**

How does the AI actually work? What assumptions are built in? What can it do? What can't it do?

## **PRACTICES**

Who uses this system? How? What labor built it? How is it constrained or enabled?

## **ACTIONS**

What happens as a result? Who benefits? Who is harmed? Does it solve the stated problem? At what cost?

# Course Structure

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Three main components form your grade:

- 1. **Class Engagement (40%)**
- 2. **Case Studies (30%)**
- 3. **Case Design (30%)**

Your letter grade will be determined by the following cutoffs:

| Earned Grade | 4.0    | 3.5            | 3.0            | 2.5            | 2.0            | 1.5            | 1.0         | 0.0    |
|--------------|--------|----------------|----------------|----------------|----------------|----------------|-------------|--------|
| Score Range  | > 3.75 | 3.74... - 3.25 | 3.24... - 2.75 | 2.74... - 2.25 | 2.24... - 1.75 | 1.74... - 1.25 | 1.24 - 0.75 | < 0.75 |

# 1. Class Engagement (40%)

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Each component counts equally towards your grade

- **Daily Class Attendance & Participation (drop 3)**
  - Being attendant is important
  - We will use lots of group work and being present for your group matters
  - *We will check for your participation daily*
- **Forum posts (weekly, drop 2) + replies (~250 words each)**
  - Posts are expected in response to prompts every week (11:59pm Sunday); APA formatting required
  - *Forum replies earn "Deeper Engagement" credit and are not required*
- **Exit tickets (after every class, ~100 words each, drop 3)**
  - Each class, you will debrief before 5pm
  - *Can also be a place to tell me how you are doing with the material or how things went today*

## 2. Case Studies (30%)

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Each component counts equally towards your grade

- **Case Studies (Google Doc Template, drop 1)**
  - Analyze AI systems using DTPA each week as a group
  - *Turn in your weekly report by Sunday 11:59pm*
- **Individual & Metacognitive Report (~250 words, drop 1)**
  - Reflect on your own learning and effort; bring your values into what we just completed
  - *Turn in your report by Sunday 11:59pm*

### 3. Case Design (30%)

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Each component counts equally towards your grade

Semester-long group project, either:

1. Design an AI application that directly addresses a major societal problem, or
2. Propose limits, safeguards, or a set of policy constraints on an existing AI system

Getting you there

- **Two scaffolded assignments (Oct 2nd & Nov 6th)**
  - Support your group's development
  - Receive and respond to feedback from instructors
- **Final showcase (draft due Dec 4th, present last week of class)**
  - Showcase is ungraded but will have class feedback to improve final turn-in
  - Presenting at showcase unlocks final turn-in
- **Final project turn in (Day of our final exam)**

# How We Grade: Labor & Standards-Based

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This is **not** a traditional grading system.

## What it means:

- Your grade reflects your effort and growth, not just "right answers"
  - *There are no right answers to what we are working on*
- Most assignments get a second chance if you don't meet the standard initially
- There's no curve, no "grading on a bell curve"

*Everyone can earn a 4.0 and we hope you will*

# How We Grade: Labor & Standards-Based

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This is **not** a traditional grading system.

## What it doesn't mean:

- "Do whatever you want and pass"
- Standards are real, and we hold to them
- You need to revise and engage with feedback

*The goal: Learn the material deeply, not just perform well on a test.*

***We have no tests anyway!***

# What Does "Meeting the Standard" Look Like?

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"In general" - specifics are available in assignments

## **Meets Standard (3.0)**

- Evidence from reliable sources; documenting sources well
- Thoughtful analysis using the DTPA framework
- Acknowledgment of complexity and multiple perspectives
- Clear and respectful communication

## **Exceeds Standard (3.5–4.0)**

- Deep research beyond assigned readings; new resources and perspectives
- Metacognitive reflection on your own learning; deep engagement
- Leadership, Exceeding Efforts, or Novel Contributions in group work



# Class Culture & Expectations

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## Here's what we need from you:

- **Show up.** Physically and mentally. Exit tickets require attendance.
- **Engage thoughtfully and respectfully.** Disagreements are ok, arguments are not.
- **Think about your position.** No one has your experience and background.
- **Disclose AI use.** If you used it tell us. We trust you.
- **Meet deadlines.** Forum posts due before class each week. Most other deadlines: Sundays 11:59pm.
- **Ask questions.** If something doesn't make sense, say so.

# Class Culture & Expectations

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## Here's what you should expect from us:

- Clear expectations and timely feedback
- A second chance on most assignments
- Support if you're struggling; space if you need it
- Respect for your time and perspectives; valuing you as a person
- Stepping in to manage issues, concerns, and disagreements

## Accommodations

We have tried to design a class that welcomes every learner

*Please submit your official accommodation paperwork and  
Speak with Danny and Jorge about your needs*

# Questions?

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When you ask a question in class and a small conversation starts



# This Week's Agenda

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1. **Course intro** (✓)
2. **Group the class!** - Get up and move around! (you're here ✓)
3. **Wednesday: Constructing Categories** - How we perceive people matters.
4. **Wednesday: Who do we make into groups?** – We'll work through the Census timeline together
  - In your assigned groups; check D2L for your first assigned group, please
  - Exchange contact information!
5. **Friday: First Case - Census Data** – Drawing on Wednesday's work; Focusing on the Data in DTPA

**Exit ticket** – Your first one is due by 5pm today on D2L.

# Activity: Making Groups - Part 1

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Break into six random groups of roughly equal size

Grab a pack of index cards

- You have **seven (7) minutes** to come up with as many things you all have in common
  - Nothing pejorative (e.g., *we all hate Drake*) or physical (e.g., *we all have red hair*)
  - Write as many as you can!
- You will have **three (3) more minutes** to identify the top 3 things y'all think the rest of the room **won't** share
  - Each of you write those 2-3 things on index cards (1 idea per card)

Keep your cards, you need them for Part 2!

# Activity: Making Groups - Part 2

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Shuffle into six *new* random groups of roughly equal size

**No more than one other person from your first group**

- Share your 3 top things; is there anything that hangs you all together?
  - Again, nothing pejorative (e.g., *we all hate Ohio State*) or physical (e.g., *we all have short hair*)
  - Decide on top 2-3 (discard unused cards; or write new cards)
- If you can't find 2-3 things, shuffle yourselves again!
  - Try hard to keep groups of equal size

**Once you 2-3 things with a new group, try to name your group.**

**You will only have ten (10) minutes. Work fast!**

# Activity: Making Groups - Wrap Up

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Share out the commonalities!

**Discussion:**

- How easy or hard was it to find groups?
- Are these groups (or categories) the same "kind of thing"?
- Are these categories truly unique?
- What does this activity tell you about categorizing people?

# Reminders for this week

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## Exit Tickets, D2L, and the Course Website

- Don't forget to complete today's Exit Ticket by 5pm (on D2L)
- Review your group assignment on D2L and connect with your group before Wednesday's class!
- Spend time before Friday (~90 minutes) going over this week's preparatory materials

## Questions?

Email Danny ([caball14@msu.edu](mailto:caball14@msu.edu)) and/or Jorge ([martjor@msu.edu](mailto:martjor@msu.edu))

**Have a great first week!**